



The Green University Campus Diagnosis Model

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Abstract

Planning for sustainable strategies and implementing sustainable practices in a university campus requires the assessment of a green level. Following a comprehensive research, it is observed that existing models are complex, are mathematically ambiguous, use an extreme amount of criteria and are applied in developed countries solely. The purpose of this paper is to propose a generic model that diagnoses the level of sustainable practices of green university campuses. The developed model, entitled the “Green University Campus Diagnosis” (GUCD) model, provides a green level of a university campus. Green indicators for university campus are defined, and criteria weights are evaluated using Shannon’s entropy method. Then, a modified version of Preference Ranking Organization Method for Enrichment Evaluation (PROMETHEE) ranks a green university campus between a “very bad” and an “excellent” green university campus. Data are collected from universities’ managers, directors and engineers in the Middle East and North Africa (MENA) region. The model is applied to four Lebanese universities. Results show that university campuses have a green level between 56.55 and 74.8%. The GUCD is a generic, simple-to-use model. The GUCD model uses well-defined mathematical Multi-Criteria Decision methodologies and an adequate number of green indicators.

Keywords Higher education · Green campus · PROMETHEE · Shannon’s entropy · Sustainability practices

Introduction

According to several researchers, Higher Education Institutes (HEI) serve as a model for communities and societies aiming to minimize greenhouse gas emissions and be eco-friendly and to improve people’s health and wellbeing (Abu Qdais et al. 2019; Choi et al. 2017; Sima et al. 2019). According to Moqbel et al. (2020), sustainable development in HEI is associated with the term “Green University” or “Green Campus”. Moezzi et al. (2014) defined green campus as the “outcome of applying sustainability strategies and practices in order to provide a better life for all university users”. In addition, Sisriany and Fatimah (2017) defined green campus as a “concept to develop sustainable practices in the university”.

Focus on a green university campus is deemed important in terms of sustainable strategy. Some universities have put initiatives in place as goals for sustainable implementation, although these goals are still fragmented and not interconnected (Vagnoni & Cavicchi 2015).

Hence, a well-defined sustainable strategy for university campuses is a must. A sustainable strategy for university campus requires the assessment of the sustainability practices, or in other words, evaluating the “green level” of the university campus.

Based on existing studies (refer to background section), it is observed that some models have been developed in order to check the sustainability practices in universities. These models lack a clear evaluation methodology, or use a huge number of criteria, or are applied in developed countries solely. Furthermore, it is noted that no model exists that evaluates mathematically a green level for a university campus.

Therefore, the main objective of this paper is to develop a model that assesses the “green level” of a university campus. The developed model follows the following methodology in order to achieve its objective: (i) identify university campus “green” indicators and respective criteria in a hierarchy

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format, (ii) evaluate the criteria importance weights, (iii) evaluate the criteria scores and (iv) use Multi-Criteria Decision Analysis (MCDA) tools in order to evaluate a Green University Campus Index.

This paper starts by presenting the background of the topic, where existing universities' sustainable practices and models are described. Next, the paper presents the Green University Campus Diagnosis (GUCD) model detailed development, and its application to selected Lebanese universities. Results are then presented and discussed. Finally, conclusions and recommendations are presented.

Background

Over the past few years, several studies on the implementation of sustainability practices and strategies are recorded in research. A thorough and comprehensive research in different databases was conducted. Table 1 summarizes the existing universities' sustainable implementation.

On one hand, several researchers have noticed that there is a lack of interdisciplinary collaboration among

sustainability practices in universities (Velazquez et al. 2005; Abdul-Azeez & Ho 2015; Sonetti et al. 2016; Azlin et al. 2016). On the other hand, integrating sustainable indicators for a green campus in a university is still an ongoing problem (Velazquez et al. 2005; Nifa et al. 2016; Sonetti et al. 2016).

It is important to note that both the Leadership in Energy and Environmental Design "LEED" (USGBC 2016) and the Building Research Establishment Environmental Assessment Method "BREEAM" (BRE 2016a, 2016b) do not have direct programs designed for university campuses, or universities in general.

On the other hand, some universities have developed assessment models in order to examine whether they are considered environmentally friendly in accordance to sustainable measures. A comprehensive research was undertaken, and Table 2 summarizes major existing models for sustainability assessment in universities.

Based on Table 2, it is observed that although some of these models are in use, some common limitations can be observed: (i) ambiguity in the evaluation methodology, (ii) most models are developed and applied in developed

Table 1 Major research in universities' sustainable practices implementation

Authors	University sustainable practices focus	Methodology	University location
Cattaneo et al. (2018)	Alternate mode of transport	Statistical analysis	Italy
Zen et al. (2016)	Minimize solid waste	Scientific math evaluation	Malaysia
Dahlawi and Sharkawy (2021)	Identification of recycled and non-recycled solid waste categories, and food products	Statistical analysis	Saudi Arabia
Abu Qdais et al. (2019)	Reduction of carbon footprint generated from energy, water consumption and solid waste	Scientific math evaluation	Jordan
Robinson et al. (2015)	Reduction of carbon emissions	Point allocation appraisal model with reality check	England
Tangwanichagapong et al. (2017)	Management of waste using 3R (Reduce, Reuse and Recycle)	Statistical analysis	Thailand
Aleixo et al. (2018)	Measurement of social, economic and environmental sustainability implementation	Statistical analysis	Portugal
Anthony Jnr (2020)	Examination of the existence and usage of sustainable practices according to green indicators	Statistical analysis	Malaysia
Nifa et al. (2014) Nifa et al. (2015)	Investigation and developing a framework for the integration of sustainable practices in the design of construction projects	Statistical analysis	Malaysia
Mat et al. (2011)	Defining environmental (air and water quality), solid waste, natural resources, economical and health sustainable criteria	-	Malaysia
Hooi et al. (2011)	Assessment of carbon footprint	Scientific math evaluation	Malaysia
Barros et al. (2020)	Application of Life Cycle Assessment (LCA) in order to assess water, electricity and waste emissions	LCA	Brazil
Lozano (2011)	Reviewing environmental, social and educational aspects, and assessing states of sustainability reporting	Graphical assessment of sustainability in universities tool	Worldwide
Genta et al. (2019)	Defining the strategies with focus on avoiding ecological footprint	Statistical analysis	Italy

Table 2 Major university sustainability assessment models

Authors/researchers	Model	Sustainable practices criteria	Methodology	Application location	Limitations
Register (2006); Eagan et al. (2008); Beringer et al. (2008); Stafford (2011); Finlay and Massey (2012)	Ecocity	Housing, Transportation, Energy, Planning, Health and social equity, Land use	Campus is considered a city replica on a smaller scale	USA	Not an assessment model
UI Green Metrics (2019)	UI Green Metrics	Setting and infrastructure, Energy and climate change, Waste, Water, Transportation, Education and research	Based on LEED and BREEAM Data collection from surveys	Indonesia and world-wide	Non-reliable in non-developed countries
People and Planet (2020)	Green League	Environment policy, Human resources, Ethical investment, Carbon management, Workers' rights, Sustainable food, Staff engagement, Student engagement, Education, Waste and recycling, Carbon reduction, Water reduction	Worth weight evaluation of each criteria	UK alone	- Non transparent and clear method of ranking - Huge number of sub criteria - All sub criteria must exist for the model to function - UK alone
Princeton Review (2020)	Princeton Review Green Rating	Quality of campus life, Eco-friendly energy, Food, Education, Waste, Transportation, Energy consumption, Greenhouse gas, Building infrastructure	Based on surveys	USA alone	- Ambiguous evaluation methodology - Uncommon model
Kirrane et al. (2020)	Sustainable Tracking Assessment and Rating System	Academics, Engagement, Operations, Planning and administration, Innovation and leadership	Based on the Association for the Advancement of Sustainability in Higher Education (AASHE)	USA alone	- Extremely huge number of sub criteria - USA only

countries only and (iii) they are complex models where an extremely huge number of criteria and sub criteria is used.

Therefore, a need to develop an assessment model in order to evaluate the “green” level of a university campus is important in order to fill in the gap in existing academic literature and practice. This paper introduces the new developed “Green University Campus Diagnosis (GUCD) model. This new model overcomes all the drawbacks of existing models:

- (i) The GUCD model is generic and can be used in developed and non-developed countries,
- (ii) The GUCD model is transparent and uses scientifically proven math models based on Multi-Criteria Decision Methodologies.
- (iii) The GUCD uses adequate number of criteria (green indicators) and sub criteria.

- (iv) The GUCD model comprises most significant sustainable strategies and practices in a university campus.

The Green University Campus Diagnosis Model

The Model Methodology

The developed “Green University Campus Diagnosis” (GUCD) model assesses the level of sustainability practices in a university campus, or in other terms the “green” level of a university campus. The GUCD model uses Multi-Criteria Decision Analysis (MCDA) tools in order to evaluate a green university campus index. This GUCD index indicates how much “green” is the campus based on certain sustainable practices indicators. The GUCD model methodology flow chart is illustrated in Fig. 1.

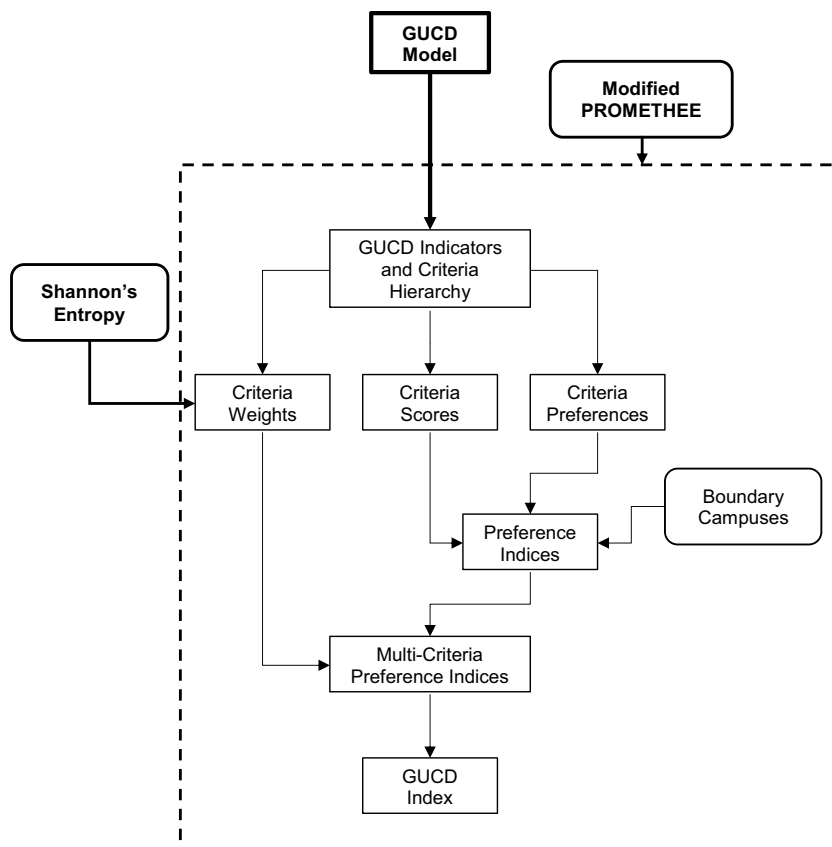
Based on the methodology illustrated in Fig. 1, the GUCD model uses the “Preference Ranking Organization Method for Enrichment Evaluation” (PROMETHEE), developed by Brans et al. (1984, 1986) in order to evaluate the criteria scores, aggregate criteria and evaluate a green index. The widespread use of this tool in many fields such as water resource management, energy planning and waste

management has pushed it to become reliable and efficient (Abdullah et al. 2019). PROMETHEE is based on comparisons and ranking of alternatives with respect to every single criterion. It allows a simple and straightforward decision-making, and its main advantage lies in its ability to compare data that do not belong to the same category and measured based on different scales (Murat et al. 2015).

The purpose of the GUCD model is to assess or diagnose a “green” level, i.e. defining a “green” index on a fixed scale. However, the PROMETHEE method outcome is a rank, and not a scaled utility. Therefore, a modification on the method is performed. Two fictitious alternatives are introduced, acting as boundary alternatives: a very bad green university campus on one hand, and an excellent one on the other. The incorporation on these two extreme cases enables PROMETHEE to rank a university between a “very bad” one and an “excellent” one, hence providing the scaled index defining the “green” level diagnosis.

In addition, the GUCD model considers that not all sustainable indicators and criteria are equally important. Hence, Shannon’s entropy (SE) is used as a method to evaluate the importance weights of the different sustainable practices criteria. The Shannon’s entropy methodology, developed by Shannon in 1948 (Shannon 1948), states that the quality of the information gathered is essential for an accurate measurement and decision making. The concept of Shannon’s

Fig. 1 The GUCD model methodology flow chart



entropy has a dominant role in information theory (Ghorbani et al. 2012). This concept has been developed to different scientific fields (Ghorbani et al. 2012). Therefore, the Shannon's entropy technique is identified as an objective method for criteria weight calculation.

The GUCD Sustainable Practices Indicators

The GUCD model uses sustainable practices indicators and respective criteria. The indicators and criteria were derived from (i) the existing researches done for universities' sustainable strategies and practices, (ii) the existing universities' sustainable assessment models and (iii) LEED and BREAM general practices. However, the GUCD model compiled these indicators and filtered the most relevant and appropriate ones for general usage and practices. Hence, experts for Higher Education Institutes in the Middle East and North Africa (MENA) region were addressed, and they chose general indicators and criteria relevant to available and/or aspired sustainable practices. The experts are directors of university campuses, academic researchers in different universities with research emphasis on sustainability. Thirty-two (32) experts were interviewed either physically or virtually. They were asked to define major indicators of what would be a "green campus". Some hints from existing research were provided as well. Delphi method was used. First interviews targeted the researchers, then a new round of interviews targeted the practitioners.

Because of this process, the GUCD method uses five major sustainable practices indicators or defined as "green" university campus indicators, as shown in Fig. 2.

Each green indicator consists of several criteria. Based on the experts' opinion, scoring input for each criterion is provided. Table 3 shows the GUCD proposed criteria definition, attribute and score, following the HEI experts' feedback.

Using Table 3, a HEI decision maker can check for each criterion the available attribute based on the sustainable practice and assign the respective score for it. The scores are then aggregated using the PROMETHEE method. PROMETHEE method allows for different scoring schemes and different criteria attributes scores and units. So, there is no need to normalize the scores or the units.

Criteria Aggregation

Criteria aggregation in PROMETHEE is done using the pseudo-criteria concept. A pseudo-criterion is an indication of how much the criterion is preferred between two alternatives. If the difference of the criterion scores ΔS_i (criterion i) is higher than a Preference Threshold (PT), there is a strict preference between the first and second alternative. If the difference is lower than a Preference Threshold (PT), a relative preference exists and a preference index (P_i) is evaluated using a defined preference function. Also, the HEI experts advised the preference function type for each criterion, and assigned the Preference Threshold (PT) for each criterion preference function. The GUCD uses two preference function types. Figure 3 illustrates the preference function types 1 and 2.

The GUCD modifies the traditional PROMETHEE method by adding boundary fictitious alternatives. A "Very Bad" (VB) and an "Excellent" (E) green university campuses alternatives are introduced. Therefore, every university campus is ranked between a "Very Bad" (U_{VB}) and an "Excellent" (U_E) green campuses. The "Very Bad" campus will have all criteria scored the lowest (S_{iVB}), while the "Excellent" campus will have the criteria scored the highest (S_{iE}). The same experts provided the "Very Bad" and the "Excellent" scores for each criterion.

Table 4 shows the GUCD criteria aggregation parameters: (i) scores of both boundary university campuses, (ii) the preference function types and (iii) the respective Preference Threshold (PT).

Comparing a certain university campus U_j to boundary university campuses (U_{VB} and U_E), the preference indices for each criterion are evaluated using Eqs. 1–6.

$$P_i(U_j; U_{VB}) = \frac{S_{ij} - S_{iVB}}{PT_i} \quad (1)$$

$$P_i(U_j; U_E) = \frac{S_{ij} - S_{iE}}{PT_i} \quad (2)$$

$$P_i(U_E; U_{VB}) = \frac{S_{iE} - S_{iVB}}{PT_i} \quad (3)$$

$$P_i(U_E; U_j) = \frac{S_{iE} - S_{ij}}{PT_i} \quad (4)$$

Fig. 2 The green university campus indicators

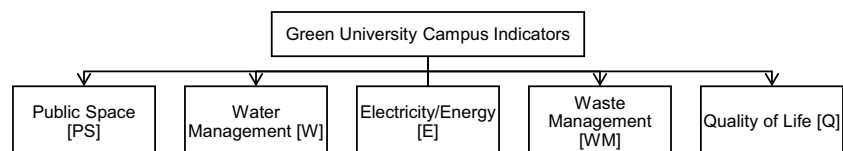


Table 3 Green indicators definitions, attributes and scoring

Green indicators	Criteria (C _i)	Criteria definition	Criteria attribute	Score S _i
Public space	Green Space Management [PS1] (C ₁)	The green space ratio is evaluated based on the total planted or green area in relation to the total campus area	Green space ratio	[%]
	Parking Lot Management [PS2] (C ₃)	Based on the LEED standardized condition which is a minimum distance of 60 m is to be respected between a parking lot and a facility	Average distance	[m]
	Bicycle and Pedestrian Access [PS3] (C ₃)	Evaluation is based on the availability of bicycle racks, pedestrian friendly routes and ramps for the disabled	None of the options available One option is available Two options are available Three options are available	1 2 3 4
Water management	Water Treatment Plant [W1] (C ₄)	Scoring is based on the availability of a water treatment plant on campus or near it, and if not, if water treatment techniques are implemented	No water treatment Water treatment processes available Water treatment plant near campus Water treatment plant on campus	1 2 3 4
	Monitoring System [W2] (C ₅)	The campus is assessed based on the availability of a monitoring system on both potable and gray water consumption	Unavailable monitoring systems Available monitoring system	0 1
	Retention Techniques [W3] (C ₆)	Assessment is based on the availability of water retention techniques on campus	Non available retention techniques Available retention techniques	0 1
Energy	Generator Type [E1] (C ₇)	Assessment is based on which alternative the campus uses to provide energy	Private generators PV panels used solely for water heating PV panels and private generators PV panels used for all electricity supply Efficiency based on the data retrieved	1 2 3 4 [%]
	PV Panel Efficiency [E2] (C ₈)	Evaluation is based on the total electricity produced by the campus via PV panels in ratio to the total electricity consumption		
	Roof Insulation [E3] (C ₉)	Evaluation is based on the availability of roof insulation or if it is possible to integrate it upon renovation	Unavailable roof insulation Possibility of integrating insulation Available roof insulation on some buildings Available roof insulation of all buildings	1 2 3 4
	Types of Lamps [E4] (C ₁₀)	Assessment is based on the type of lamps used due to their impact on the electricity consumption	Incandescent lamps used solely LED lamps solely LED and natural lighting LED, HIDs and natural lighting	1 2 3 4

Table 3 (continued)

Green indicators	Criteria (C_i)	Criteria definition	Criteria attribute	Score S_i
Waste management	Number of Shifts [WM1] (C_{11})	Assessment is based on the number of workers per shift per day, the more the better	Unavailable staff daily	1
			1 worker per day	2
			2–3 workers per day	3
			More than 3 workers per day	4
Quality of campus life	Discharge Station Proximity [WM1] (C_{12})	Assessment is based on the distance or location of the discharge station to the campus	Discharge station near facilities	0
			Discharge station far from facilities or off campus	1
	Recycling Efficiency [WM2] (C_{13})	The recycling ratio is based on the amount recycled over the among generated	Efficiency based on the data retrieved	[%]
	Construction Waste [WM3] (C_{14})	Assessment is based on the availability of storage stations along with respective safety measures undertaken	Absence of construction waste disposal for re-use	0
			Available storage of construction waste for re-use	1
	Smoking Area [Q1] (C_{15})	Assessment is based on the respected distance for designated smoking areas, generally, 15 m away from any building entrance	Distance between smoking area and a facility entrance is less than 7.5 m or smoking allowed all over campus	0
			Distance is greater than 7.5 m or no smoking allowed on campus	1
	Outdoor Living Space [Q2] (C_{16})	Assessment is based on the availability of an outdoor living space surrounding certain faculties	No outdoor living space surrounding facilities	0
			Available outdoor living space surrounding the facility	1

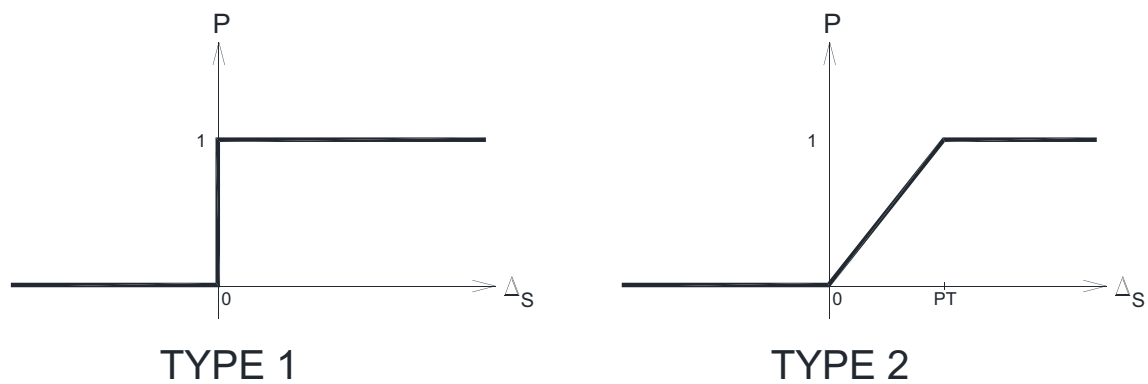


Fig. 3 Preference function types

Table 4 GUCD criteria aggregation parameters

Criteria C_i	“Very Bad” University (U_{VB}) scores S_{iVB}	“Excellent” University (U_E) scores S_{iE}	Preference threshold PT_i	Preference function type
PS1 (C_1)	0%	100%	90%	2
PS2 (C_2)	10 m	70 m	60 m	2
PS4 (C_3)	1	4	4	2
W1 (C_4)	1	4	4	2
W2 (C_5)	0	1	0	1
W3 (C_6)	0	1	0	1
E1 (C_7)	1	4	4	2
E2 (C_8)	0%	100%	70%	2
E3 (C_9)	1	4	4	2
E4 (C_{10})	1	4	4	2
WM1 (C_{11})	1	4	4	2
WM2 (C_{12})	0	1	0	1
WM3 (C_{13})	0%	100%	80%	2
WM4 (C_{14})	0	1	0	1
Q1 (C_{15})	0	1	0	1
Q2 (C_{16})	0	1	0	1

$$P_i(U_{VB}; U_E) = \frac{S_{iVB} - S_{iE}}{PT_i} \quad (5)$$

$$P_i(U_{VB}; U_j) = \frac{S_{iVB} - S_{ij}}{PT_i} \quad (6)$$

where: i = criteria and j = university campuses

Criteria Weights

The criteria weights evaluation is based on the Shannon’s entropy method. Shannon’s entropy is found to be the best method of weight evaluation for the GUCD model because it does not require lot of data collection. The method uses the

criteria scores in order to evaluate the weights. Using the criteria scores, a normalized score is evaluated as defined in Eq. 7.

$$p_{ij} = \frac{S_{ij}}{\sum_{i=0}^n S_{ij}} \quad (7)$$

where: S_{ij} = score of criteria i for university j

n = number of criteria = 20

Next, the entropy is evaluated using Eq. 8.

$$E_i = -\frac{1}{\ln(m)} \times \sum_{j=0}^m (p_{ij} \times \ln(p_{ij})) \quad (8)$$

where: m = number of alternatives = 3

Next, the degree of diversification is evaluated using Eq. 9.

$$d_i = 1 - E_i \quad (9)$$

Finally, the weight is evaluated using Eq. 10.

$$w_i = \frac{d_i}{\sum_{i=0}^n d_i} \quad (10)$$

where: w_i = weight of criterion i

Multi-Criteria Preference Index and Flows

The GUCD model follows the PROMETHEE method in order to evaluate a multi-criteria preference index “ Π ”. The multi-preference index is the weighted sum of the preference indices “ P_i ” evaluated in the previous steps. Hence, a multi-criteria preference index is calculated for each set of two universities (alternatives) as defined in Eqs. 11–16:

$$\Pi(U_j; U_{VB}) = \sum_{i=1}^n w_i \times P_i(U_j; U_{VB}) \quad (11)$$

$$\Pi(U_j; U_E) = \sum_{i=1}^n w_i \times P_i(U_j; U_E) \quad (12)$$

$$\Pi(U_E; U_{VB}) = \sum_{i=1}^n w_i \times P_i(U_E; U_{VB}) \quad (13)$$

$$\Pi(U_E; U_j) = \sum_{i=1}^n w_i \times P_i(U_E; U_j) \quad (14)$$

$$\Pi(U_{VB}; U_E) = \sum_{i=1}^n w_i \times P_i(U_{VB}; U_E) \quad (15)$$

$$\Pi(U_{VB}; U_j) = \sum_{i=1}^n w_i \times P_i(U_{VB}; U_j) \quad (16)$$

According to PROMETHEE, the measure of strength of a specific alternative (university campus in the GUCD model) is defined as the sum of multi-criteria preference indices of this particular alternative compared to the rest, while the measure of weakness of a specific alternative is defined as the sum of multi-criteria preference indices of all alternatives compared to this specific alternative. Both measures are defined as the positive flow (strength) and the negative flow (weakness) respectively. Thus, the net flow “ ϕ^{NET} ” is the difference of the positive flow and the negative flow. The net flows of the GUCD model are defined in Eqs. 17–19:

$$\phi^{NET}(U_j) = \{\Pi(U_j; U_{VB}) + \Pi(U_j; U_E)\} - \Pi(U_{VB}; U_j) - \Pi(U_E; U_j) \quad (17)$$

$$\phi^{NET}(U_{VB}) = \{\Pi(U_{VB}; U_j) + \Pi(U_{VB}; U_E)\} - \Pi(U_j; U_{VB}) - \Pi(U_E; U_{VB}) \quad (18)$$

$$\phi^{NET}(U_E) = \{\Pi(U_E; U_j) + \Pi(U_E; U_{VB})\} - \Pi(U_j; U_E) - \Pi(U_{VB}; U_E) \quad (19)$$

The GUCD Index

The GUCD’s green score of a university U_j is a well-defined measure (net flow) between the net flow of a very bad university campus and the net flow of an excellent university campus. The GUCD green score is defined as the GUC_{INDEX} and evaluated using Eq. 20:

$$GUC_{INDEX}(\%) = \frac{\phi^{NET}(U_j) - \phi^{NET}(U_{VB})}{\phi^{NET}(U_E) - \phi^{NET}(U_{VB})} \times 100 \quad (20)$$

The Green University Campus Index GUC_{INDEX} indicates how far the U_j is from the U_{VB} (ranging between U_{VB} and U_E).

Sensitivity Analysis and the GUCD Index Scale

In order to identify a scale of “green” campus for a specific university U_j , sensitivity analysis is performed. In addition to the extreme boundaries U_{VB} and U_E , trial universities were used such as U_{VL} (Very Low university), U_L (Low university), U_{AVE} (Average university), U_G (Good university) and U_{VG} (Very Good university). The GUC_{INDEX} of the trial universities are evaluated as shown in Fig. 4.

Observing the sensitivity analysis shown in Fig. 4, a cluster of scales of the GUC_{INDEX} with a respective “green level” is evaluated and shown in Table 5.

Implementation of the GUCD Model and Results

Data Collection

The GUCD model is implemented to available Lebanese universities. Four universities similar in (i) campus size, (ii)

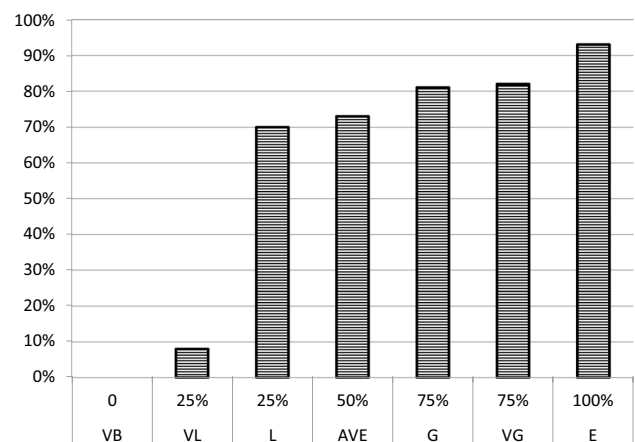


Fig. 4 Trial universities GUC_{INDEX} results

Table 5 The GUC_{INDEX} scale

GUC _{INDEX} range	University Campus Green level
100 to 80%	Very satisfactory “dark green”
79 to 70%	Satisfactory “regular green” or “green”
69 to 40%	Low satisfactory “light green”
39% and below	Not satisfactory “yellow”

student and faculty numbers, (iii) method of teaching, (iv) amount of published research and (v) method of administration are chosen. Campus directors (major practitioners) were contacted in these universities, and the criteria scores were assigned for each university campus. Table 6 shows the criteria scores of the different universities U_1 to U_4 (U_j ; $j = 1$ to 4).

Results and Discussion

As per the GUCD model methodology, a “Very Bad” U_{VB} and an “Excellent” U_E are introduced. Hence, each university is compared independently to U_{VB} and U_E . The U_{VB} and U_E scores are shown in Table 4.

Table 7 shows the different criteria weights.

Using the criteria weights, the indicators’ weights are evaluated and illustrated in Fig. 5.

It is observed that waste management has the highest weight, followed by public space and energy indicators,

while water management and quality of life are the least important indicators. However, this observation cannot be generalized by all means.

The GUC_{INDEX} of each university is evaluated using the data collected for scores and criteria weights. Since each university is compared to a “Very Bad” and an “Excellent” university independently.

Table 8 shows the GUC_{INDEX} results for all four universities, in addition to each indicator green campus index.

It can be observed that the four universities have “Low Satisfactory” green level overall, or “Light Green”. This observation resulted from the aggregation of the different criteria. But looking at each indicator green index, the four universities fail in the public space and waste management green levels, while they are satisfactory in the energy, water management and public space. Whereas, three out of four universities have a highly green level in the quality of life indicator, while one university (U_4) fails.

Analyzing the results, it is clear that these universities have no plan for major green policies for their campuses. However, some of the indicators are considered green, while some need proper green planning and implementation.

Conclusion

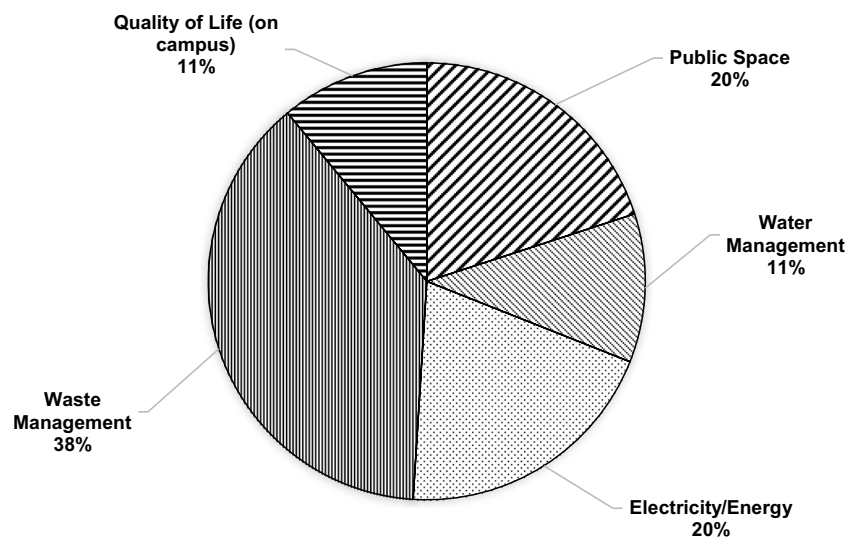
Evaluating the level of a green university campus is a very important task in the sustainable planning on one hand and checking the sustainability practices on the other. Based on existing studies (refer to the background section), it is observed that some models have been developed in order to

Table 6 Universities’ criteria scores

Criteria C_i	U_1 S_{i1}	U_2 S_{i2}	U_3 S_{i3}	U_4 S_{i4}
PS1 (C_1)	75.36	4.1	13.46	35
PS2 (C_2)	10	10	10	10
PS3 (C_3)	4	4	4	4
W1 (C_4)	4	2	4	4
W2 (C_5)	1	1	1	1
W3 (C_6)	1	1	1	1
E1 (C_7)	3	3	3	3
E2 (C_8)	1.15	3.94	2.92	2.15
E3 (C_9)	3	1	3	3
E4 (C_{10})	4	2	4	3
WM1 (C_{11})	4	4	4	4
WM2 (C_{12})	1	0	0	1
WM3 (C_{13})	23	0	0	0
WM4 (C_{14})	0	0	1	1
Q1 (C_{15})	1	1	1	0
Q2 (C_{16})	1	1	1	1

Table 7 Criteria weights from Shannon’s entropy

Criteria C_i	Criteria weights w_i
PS1 (C_1)	8.38%
PS2 (C_2)	5.82%
PS3 (C_3)	5.71%
W1 (C_4)	1.54%
W2 (C_5)	4.60%
W3 (C_6)	4.60%
E1 (C_7)	2.56%
E2 (C_8)	12.36%
E3 (C_9)	2.84%
E4 (C_{10})	2.67%
WM1 (C_{11})	2.67%
WM2 (C_{12})	11.38%
WM3 (C_{13})	12.07%
WM4 (C_{14})	11.72%
Q1 (C_{15})	6.46%
Q2 (C_{16})	4.60%

Fig. 5 The GUDC indicator weights**Table 8** The GUDC application result

	U ₁	U ₂	U ₃	U ₄
GUC _{INDEX}	66% <i>Low satisfactory</i>	44% <i>Low satisfactory</i>	63% <i>Low satisfactory</i>	69% <i>Low satisfactory</i>
PS _{INDEX}	42%	29%	28%	36%
W _{INDEX}	100%	88%	100%	100%
E _{INDEX}	88%	58%	89%	84%
WM _{INDEX}	46%	20%	54%	99%
Q _{INDEX}	100%	100%	100%	27%

check the sustainability practices in universities. These models lack a clear evaluation methodology, or use a huge number of criteria, or are applied in developed countries solely. Furthermore, it is noted that no model exists that evaluates mathematically a green level for a university campus. This paper overcomes these gaps, by presenting the Green University Campus Diagnosis (GUCD) model. This model evaluates a green index level of a university campus. The model considers five green indicators: public space, water management, energy, waste management and quality of life (on campus). For each indicator, several criteria are defined. The GUCD model evaluates the criteria scores, then evaluates the criteria weights using the Shannon's entropy method. Then, the GUCD model uses a modified version of an outranking multi-criteria decision method, PROMETHEE, in order to evaluate a Green University Campus Index (GUCD_{INDEX}). The GUCD model is applied to four Lebanese universities. All university campuses have shown low satisfactory green levels. Thus, proper green policies must be well set, and green planning and funding must be performed.

The GUCD model solves the ambiguity present in existing university rating tools; hence, the methodology is clear and simple. Moreover, currently the GUCD used expert's

opinions from the MENA region; however, the model needs to be applied to several universities worldwide in order to validate its results.

Author Contribution Both authors, Nabil Semaan and Elie Khoderchah, wrote the main manuscript, prepared all figures and tables and reviewed the whole document.

Data Availability The data sets generated and analyzed during the current study are not publicly available due confidentiality concerns, but are available from the corresponding author on reasonable request.

Declarations

Conflict of Interest The authors declare no competing interests.

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